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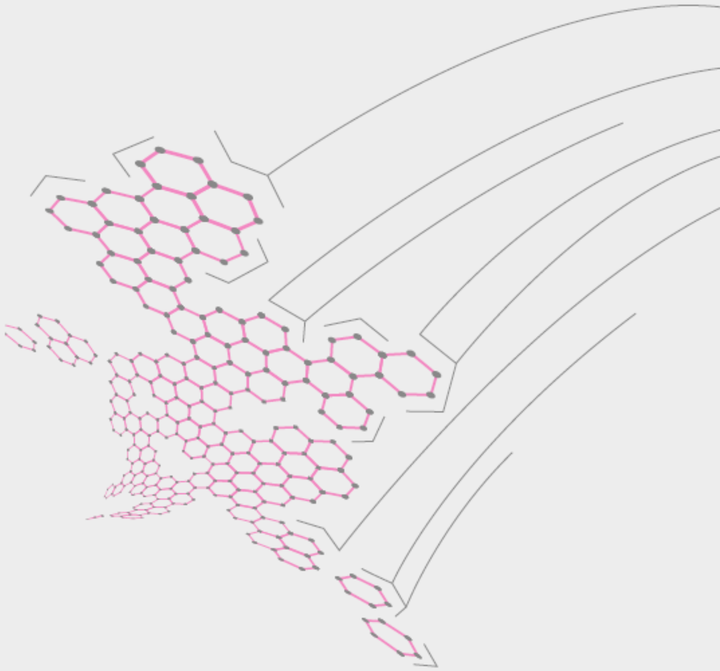
Calculus A: Lecture 5

Curves and Vector Functions

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Parametrizations of Plane Curves

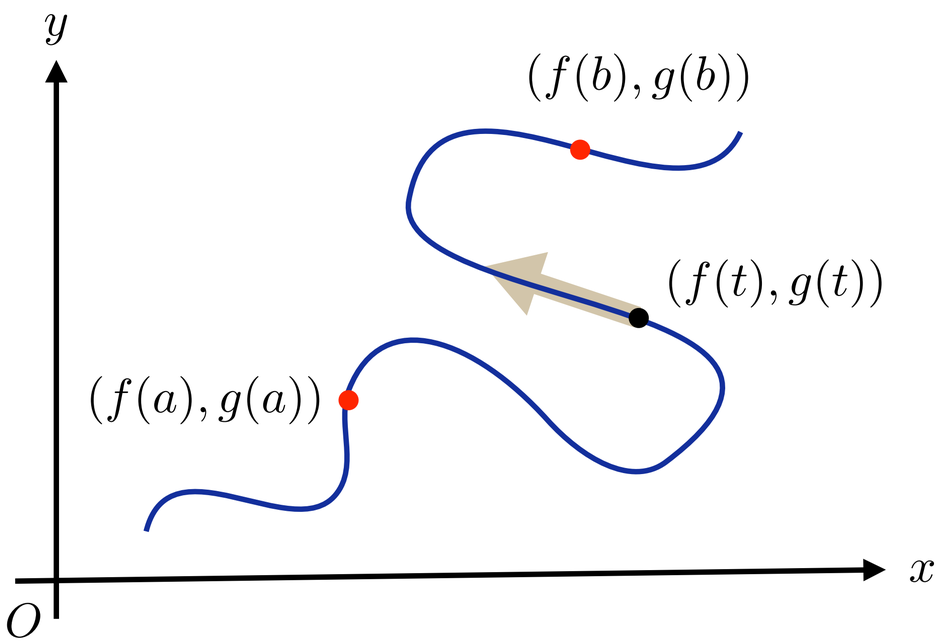
Parametric equations let us describe curves that cannot be represented as functions $y = f(x)$.

Definition: If x and y are given as functions

$$x = f(t), \quad y = g(t)$$

over an interval I of t-values, then the set of points $(x, y) = (f(t), g(t))$ defined by these equations is a parametric curve. The equations are parametric equations for the curve.

- t = parameter, with domain I (the parameter interval).
- If $I = [a, b]$:
  - Initial point: $(f(a), g(a))$
  - Terminal point: $(f(b), g(b))$
- Parametric equations + interval = a parametrization.
- A curve can have different parametrizations.



Parametrizations of Plane Curves

Example: Graph the parametric curve:

$$x = \cos t, \quad y = \sin t, \quad 0 \le t \le 2\pi$$

Find the relationship between x and y and sketch the curve.

Solution. From the Pythagorean identity $\cos^2 t + \sin^2 t = 1$, we obtain

$$x^2 + y^2 = 1.$$

Thus the curve is the unit circle centered at the origin. The parameter t increases counterclockwise, starting at $(1, 0)$ when $t = 0$.

Example:

(a) Sketch the curve defined by the parametric equations

$$x = t^2, \quad y = t + 1, \quad -\infty < t < \infty.$$

(b) Identify geometrically the curve by eliminating the parameter t and obtaining an algebraic equation in x and y.

Parametrizations of Plane Curves

Example: (multiple parametrizations of a plane curve) Consider the unit circle:

$$x^2 + y^2 = 1.$$

- **Parametrization 1** (counterclockwise, standard):

$$\mathbf{r}_1(t) = (\cos t, \sin t), \quad 0 \leq t \leq 2\pi$$

- **Parametrization 2** (clockwise):

$$\mathbf{r}_2(u) = (\cos(-u), \sin(-u)) = (\cos u, -\sin u), \quad 0 \leq u \leq 2\pi$$

- **Parametrization 3** (slower traversal):

$$\mathbf{r}_3(s) = (\cos(\tfrac{s}{2}), \sin(\tfrac{s}{2})), \quad 0 \leq s \leq 4\pi$$

All three describe the same curve, the unit circle

Calculus with Parametric Curves

- A curve $x = f(t),\ y = g(t)$ is differentiable if f and g are differentiable. (more details later)
- At points where y is differentiable as a function of x:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$ (chain rule)

Parametric Formula for $\frac{dy}{dx}$: If all three derivatives exist and $\frac{dx}{dt} \neq 0$, we have

$$\frac{dy}{dx} = \frac{dy}{dt}\left(\frac{dx}{dt}\right)^{-1}$$

If y is twice differentiable as a function of x, we can compute:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{\dfrac{d}{dt}\left(\dfrac{dy}{dx}\right)}{\dfrac{dx}{dt}}$$

at all points where $\frac{dx}{dt} \neq 0$

Length of a Parametrically Defined Curve

Example: Find the tangent to the curve

$$x = \sec t, \quad y = \tan t, \quad -\frac{\pi}{2} < t < \frac{\pi}{2}$$

$\left(\sec t = \dfrac{1}{\cos t}, \quad \tan t = \dfrac{\sin t}{\cos t}\right)$

$\left(\csc t = \dfrac{1}{\sin t}\right)$

at the point $(\sqrt{2}, 1)$, where $t = \pi/4$.

Solution. We have

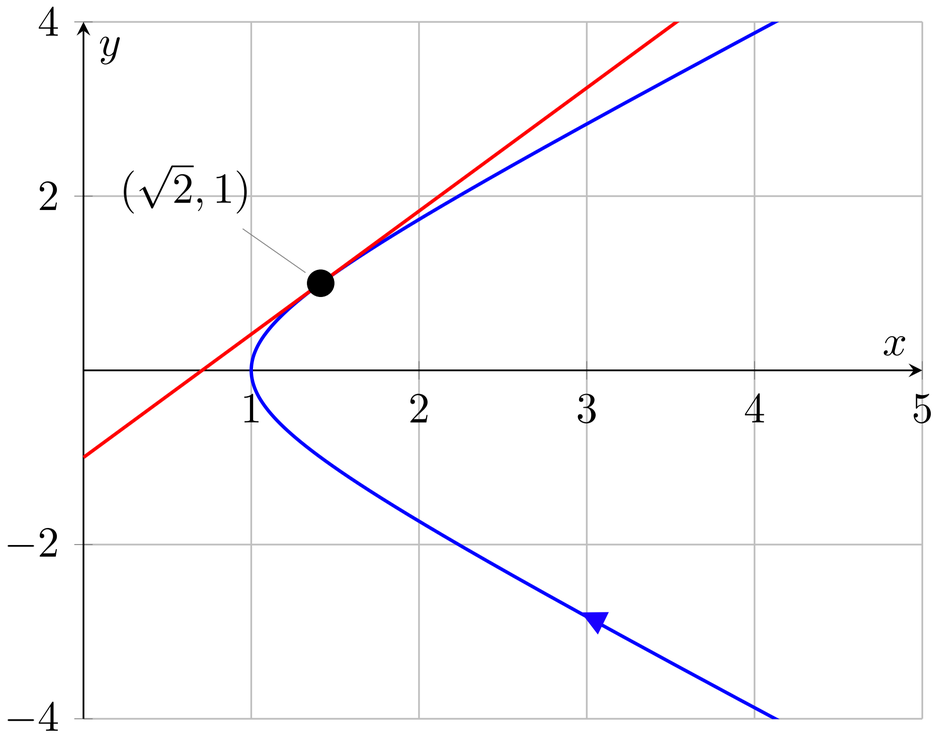
$$\frac{dx}{dt} = \sec t \tan t, \quad \frac{dy}{dt} = \sec^2 t$$

so

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\sec^2 t}{\sec t \tan t} = \frac{\sec t}{\tan t} = \csc t.$$

At $t = \pi/4$, slope $= \csc(\pi/4) = \sqrt{2}$. Equation of tangent:

$$y - 1 = \sqrt{2}(x - \sqrt{2}) \quad \Longrightarrow \quad y = \sqrt{2}x - 1.$$



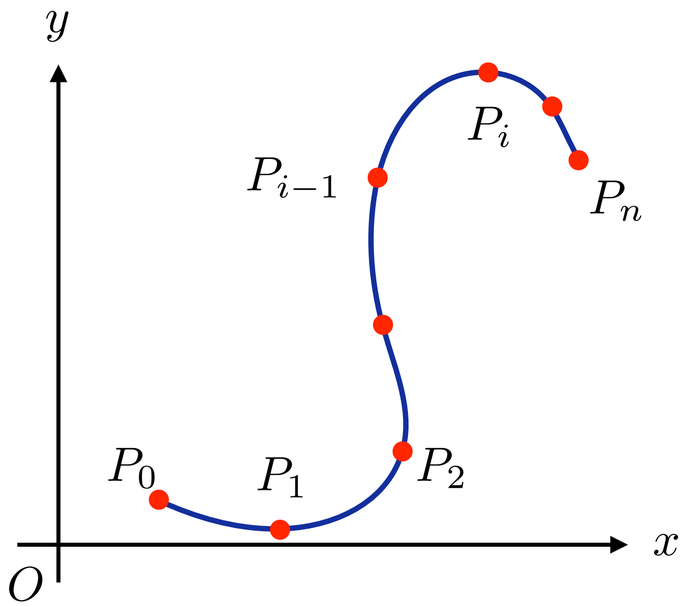
Length of a Parametrically Defined Curve

Let a curve C be parametrized by $x = f(t), \quad y = g(t), \quad a \le t \le b.$

- Divide $[a, b]$ into n subintervals $a = t_0 < t_1 < \cdots < t_n = b.$
- Approximate the curve by connecting successive points $P_i = (f(t_i), g(t_i)).$
- The length of each segment is $\Delta L_i = \sqrt{[f(t_i) - f(t_{i-1})]^2 + [g(t_i) - g(t_{i-1})]^2}.$
- Using the definition of the derivative, for small $\Delta t_i = t_i - t_{i-1}$:

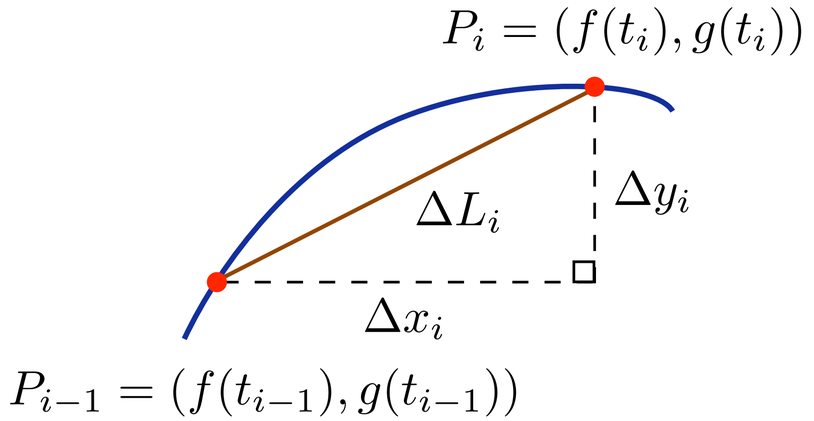
$$f(t_i) - f(t_{i-1}) \approx f'(t_{i-1})\Delta t_i, \quad g(t_i) - g(t_{i-1}) \approx g'(t_{i-1})\Delta t_i.$$

  Therefore, $\Delta L_i \approx \sqrt{[f'(t_{i-1})]^2 + [g'(t_{i-1})]^2}\,\Delta t_i.$



- Summing over all subintervals gives a Riemann sum:

$$\sum_{i=1}^{n} \sqrt{[f'(t_{i-1})]^2 + [g'(t_{i-1})]^2}\,\Delta t_i.$$



- Taking the limit as $\max \Delta t_i \to 0$ yields the definite integral:

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2}\,dt.$$

Length of a Parametrically Defined Curve

Definition: If a curve C is defined parametrically by $x = f(t)$ and $y = g(t), a \le t \le b$, where f' and g' are continuous and not simultaneously zero on $[a, b]$, and C is traversed exactly once as t increases from $t = a$ to $t = b$, then the length of C is the definite integral

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2}\, dt$$

Example: Using the definition, find the length of the circle of radius $r > 0$ defined parametrically by

$$x = r\cos t, \quad y = r\sin t, \quad 0 \le t \le 2\pi.$$

Solution. $\dfrac{dx}{dt} = -r\sin t$, $\dfrac{dy}{dt} = r\cos t$, hence

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{r^2\sin^2 t + r^2\cos^2 t} = \sqrt{r^2} = r$$

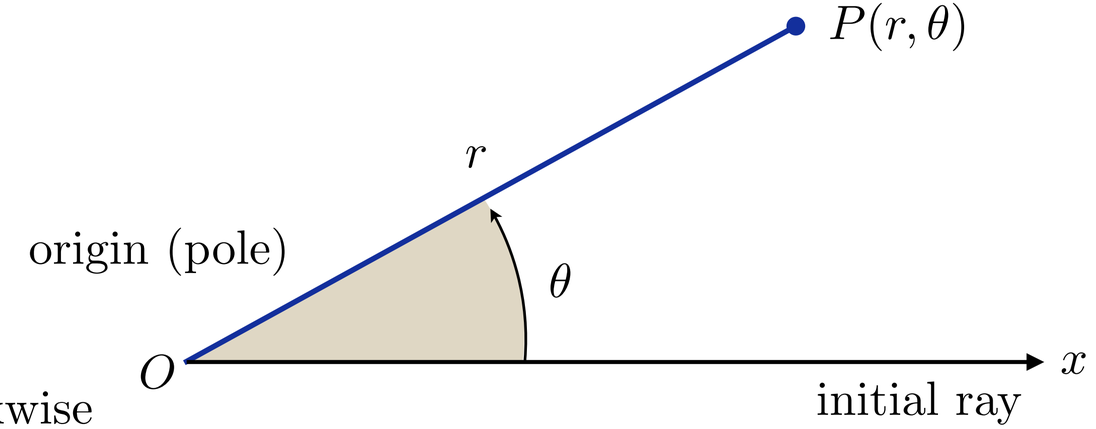
Therefore

$$L = \int_0^{2\pi} r\, dt = rt\Big|_0^{2\pi} = 2\pi r$$

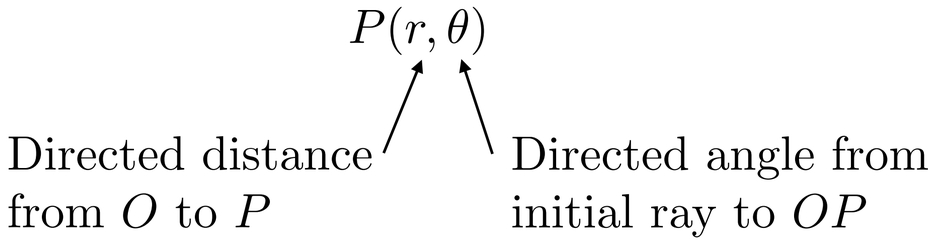
so the circle's length is $2\pi r$.

Polar Coordinates

- Fix an origin O (the **pole**) and an initial ray from O.
- A point P is described by polar coordinates (r, θ):
  - r = directed distance from O to P
  - θ = directed angle from the initial ray to OP
- $\theta > 0$: measured counterclockwise; $\theta < 0$: measured clockwise
- Polar coordinates are **not unique**:
  - A point has one Cartesian pair (x, y)
  - But infinitely many polar pairs (r, θ)



Polar Coordinates:



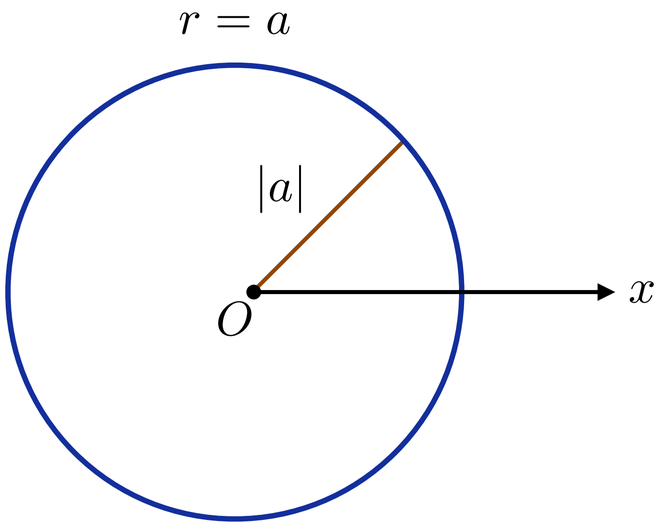
Example: the point 2 units from O on $\theta = \pi/6$

$$(r, \theta) = \left(2, \frac{\pi}{6}\right) \quad \text{or} \quad \left(2, -\frac{11\pi}{6}\right)$$

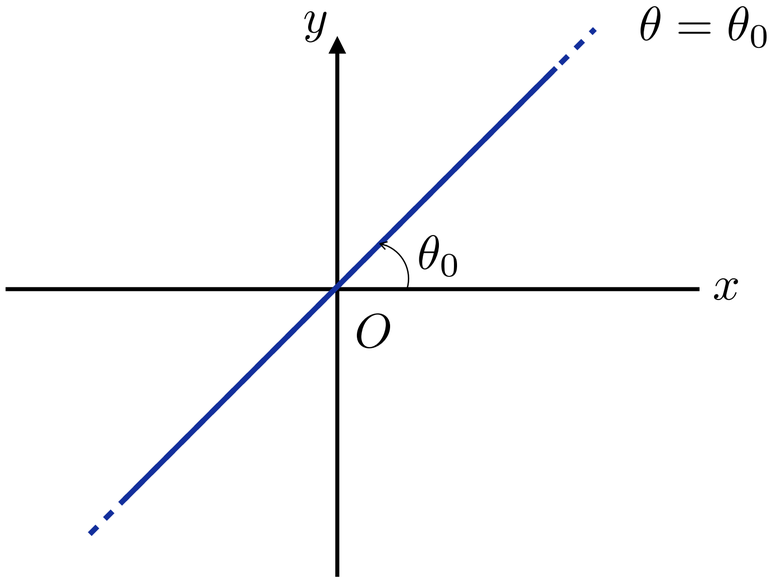
Allowing negative r: $\left(2\frac{7\pi}{6}\right) \equiv \left(-2, \frac{\pi}{6}\right)$

Polar Equations and Graphs

- If $r = a \neq 0$ (constant):
  - $P(r, \theta)$ lies $|a|$ units from O
  - As θ varies over 2π, P traces a circle of radius $|a|$ centered at O
- If $\theta = \theta_0$ (constant):
  - As r varies over $(-\infty, \infty)$, P traces a line through O
  - Line makes angle θ_0 with the initial ray



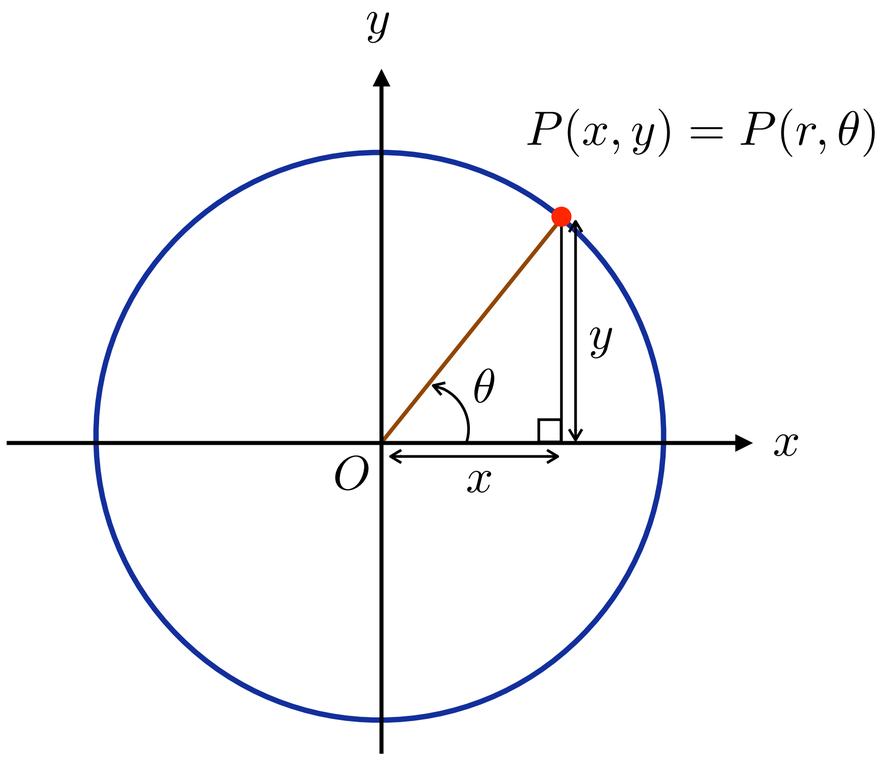
Equation	**Graph**
$r = a$	Circle of radius $\|a\|$ centered at O
$\theta = \theta_0$	Line through O making angle θ_0 with the initial ray



Relating Polar and Cartesian Coordinates

Equations Relating Polar and Cartesian Coordinates:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad r^2 = x^2 + y^2, \quad \tan\theta = \frac{y}{x}$$



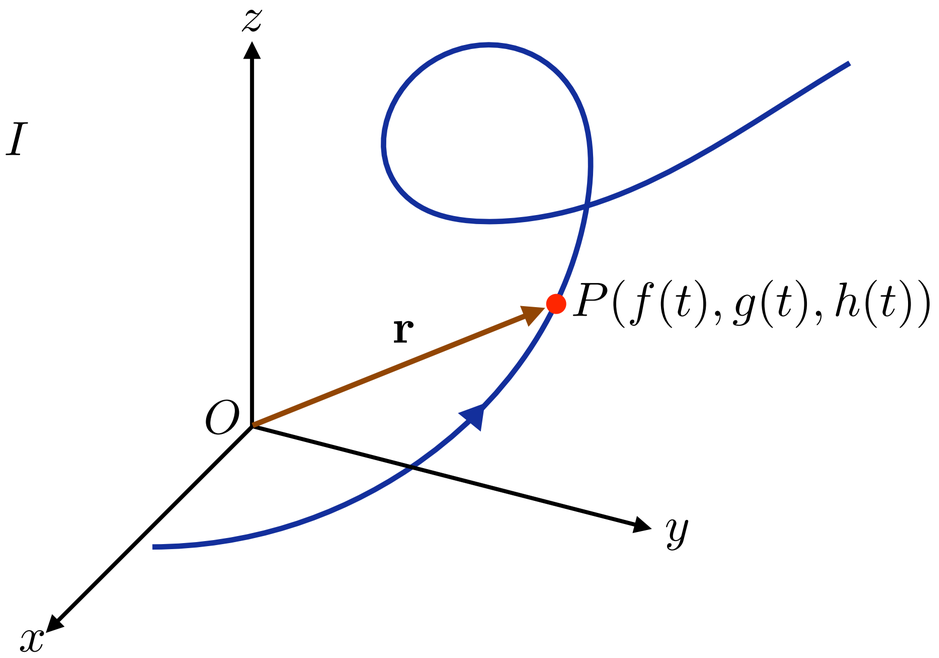
Example: Find the quivalent equations expressed in terms of Cartesian coordinates:

Polar equation	**Cartesian equivalent**
$r\cos\theta = 2$	$x = 2$
$r^2\cos\theta\sin\theta = 4$	$xy = 4$
$r^2\cos^2\theta - r^2\sin^2\theta = 1$	$x^2 - y^2 = 1$
$r = 1 + 2r\cos\theta$	$y^2 - 3x^2 - 4x - 1 = 0$
$r = 1 - \cos\theta$	$x^4 + y^4 + 2x^2y^2 + 2x^3 + 2xy^2 - y^2 = 0$
$r = \frac{\frac{16}{5}}{1+\frac{3}{5}\cos\theta}$	$\frac{(x+3)^2}{25} + \frac{y^2}{16} = 1$

Curves in Space and Their Tangents

- A particle moving through space during interval I has coordinates:

$$x = f(t), \quad y = g(t), \quad z = h(t), \quad t \in I$$



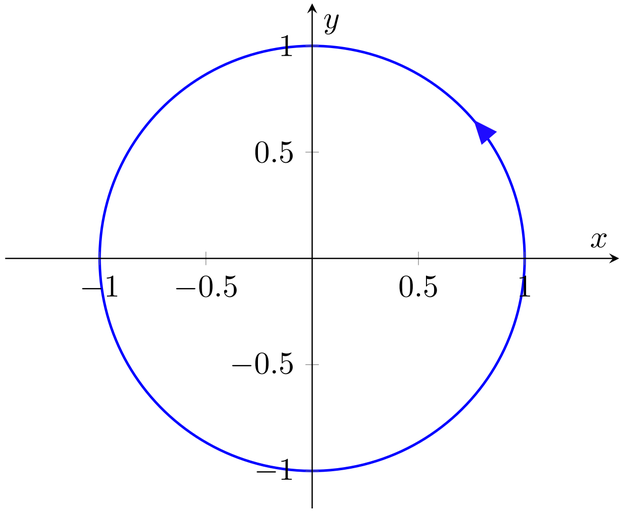
- The points $(x, y, z) = (f(t), g(t), h(t))$ form the particle's **path**.
- Vector form: position vector

$$\mathbf{r}(t) = \overrightarrow{OP} = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

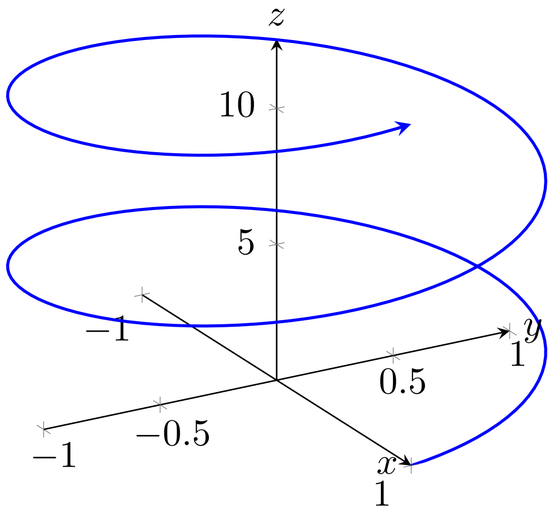
- Component functions f, g, h are scalar functions of t.

Example: Graph the vector function: $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$.

Top-down view of $\mathbf{r}(t)$



... a helix in 3D



Curves in Space and Their Tangents

Definition: A **vector function** of a real variable t is a function that assigns a vector in space to each value of t in its domain. If the vector lies in three-dimensional space, we can write

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k},$$

where $f(t)$, $g(t)$, and $h(t)$ are real-valued functions called the **components** of $\mathbf{r}(t)$. The **domain** of $\mathbf{r}(t)$ is the set of all t for which all its components are defined.

- **Vector function**: assigns a vector to each element of a domain (here, an interval I).
- Vector functions generalize to:
  - Space curves (domain = interval of $\mathbb{R}$)
  - Surfaces (domain = region in $\mathbb{R}^2$)
  - Vector fields (flow of fluids, gravitational/electromagnetic fields)

  } Calculus B
- Domain of a vector function = common domain of its components.

Limits and Continuity

For a vector-valued function

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k},$$

the concepts of limit, continuity, and differentiability can all be understood in terms of its scalar component functions $f(t)$, $g(t)$, and $h(t)$.

Definition: Let $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ be a vector function with domain D, and let $\mathbf{L} = L_1\mathbf{i} + L_2\mathbf{j} + L_3\mathbf{k}$. We say that $\mathbf{r}(t)$ has limit $\mathbf{L}$ as $t \to t_0$, written

$$\lim_{t \to t_0} \mathbf{r}(t) = \mathbf{L},$$

if and only if each of its components has a limit at t_0:

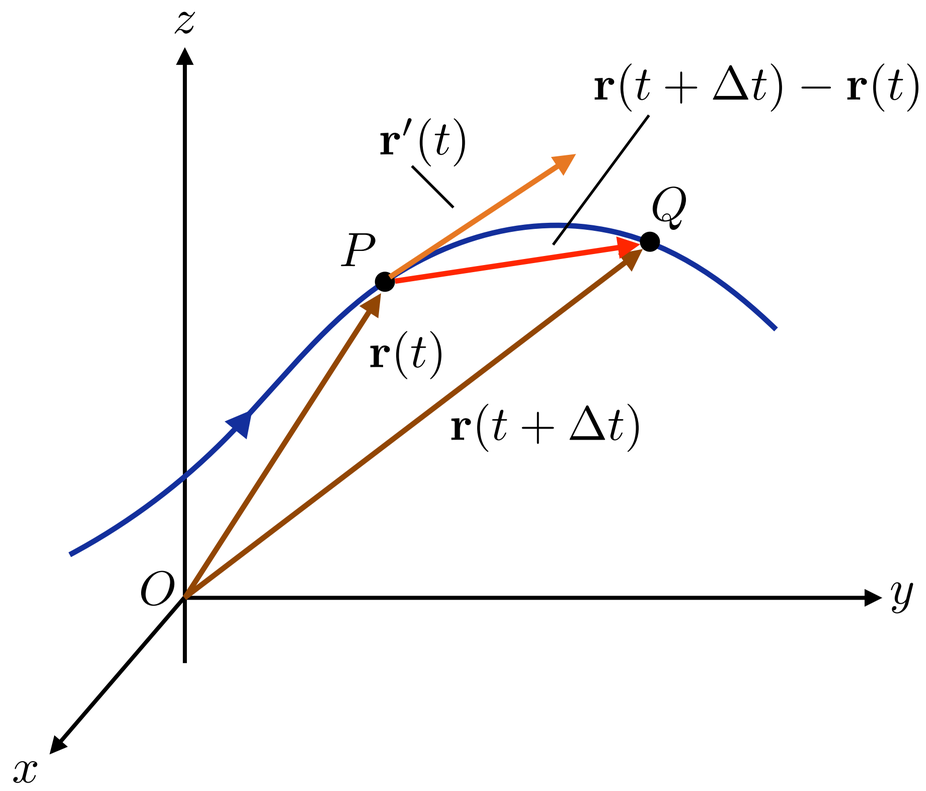
$$\lim_{t \to t_0} f(t) = L_1, \quad \lim_{t \to t_0} g(t) = L_2, \quad \lim_{t \to t_0} h(t) = L_3.$$

Definition: A vector function $\mathbf{r}(t)$ is **continuous** at a point $t = t_0$ in its domain if

$$\lim_{t \to t_0} \mathbf{r}(t) = \mathbf{r}(t_0).$$

The function is **continuous** if it is continuous at every point in its domain.

Derivative



$$\frac{\mathbf{r}(t+\Delta t)-\mathbf{r}(t)}{\Delta t} = \left(\frac{f(t+\Delta t)-f(t)}{\Delta t}\right)\mathbf{i} + \left(\frac{g(t+\Delta t)-g(t)}{\Delta t}\right)\mathbf{j} + \left(\frac{h(t+\Delta t)-h(t)}{\Delta t}\right)\mathbf{k}$$

$$\longrightarrow \underbrace{f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k}}_{\mathbf{r}'(t)} \text{ as } \Delta t \to 0$$

Definition: The vector function $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ has a derivative (is differentiable) at t if f, g, and h have derivatives at t. The derivative is the vector function

$$\mathbf{r}'(t) = \frac{d\mathbf{r}}{dt} = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t+\Delta t)-\mathbf{r}(t)}{\Delta t} = \frac{df}{dt}\mathbf{i} + \frac{dg}{dt}\mathbf{j} + \frac{dh}{dt}\mathbf{k}$$

Velocity, Speed, and Acceleration

Definition: Let $\mathbf{r}(t)$ be the position vector of a particle moving along a smooth curve in space. Then

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}$$

is the particle's **velocity vector**, tangent to the curve. At any time t:

- The **direction** of $\mathbf{v}$ is the direction of motion.
- The **magnitude** of $\mathbf{v}$ is the **speed**:

$$\text{Speed} = |\mathbf{v}|.$$

- The derivative

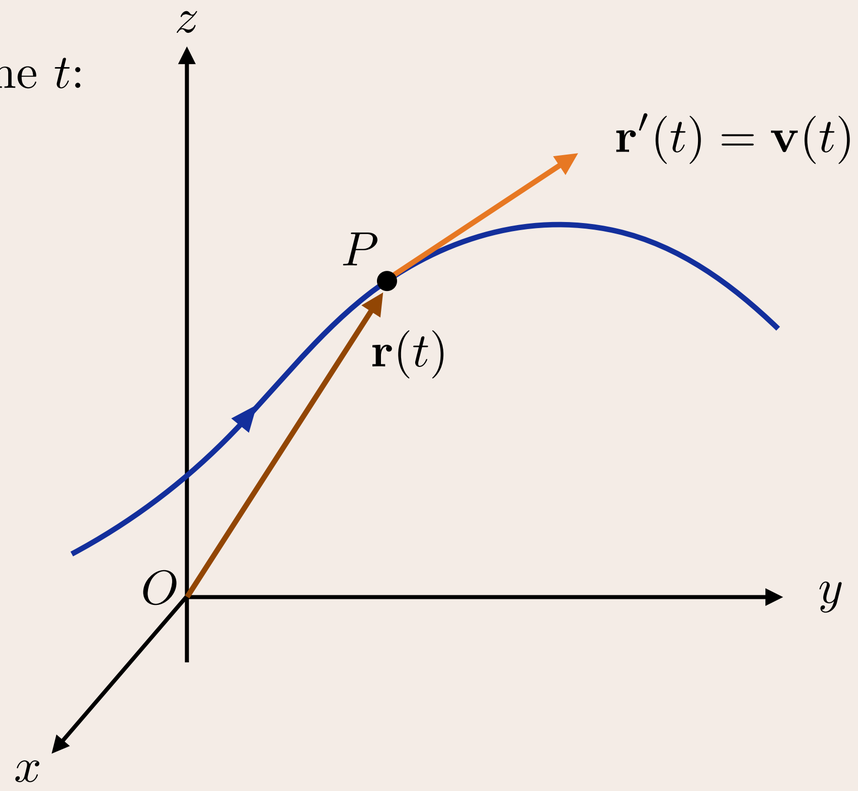
$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}$$

  is the **acceleration vector**.

- The unit vector

$$\frac{\mathbf{v}}{|\mathbf{v}|}$$

  gives the direction of motion at time t.



Derivative of Vector Functions

Differentiation Rules for Vector Functions: Let $\mathbf{u}$ and $\mathbf{v}$ be differentiable vector functions of t, $\mathbf{C}$ a constant vector, c any scalar, and f any differentiable scalar function.

1. $\dfrac{d}{dt}\mathbf{C} = \mathbf{0}$ (Constant)
2. $\dfrac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$ (Scalar multiple)

   $\dfrac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$
3. $\dfrac{d}{dt}[\mathbf{u} + \mathbf{v}] = \mathbf{u}' + \mathbf{v}'$ (Sum)
4. $\dfrac{d}{dt}[\mathbf{u} - \mathbf{v}] = \mathbf{u}' - \mathbf{v}'$ (Difference)
5. $\dfrac{d}{dt}[\mathbf{u} \cdot \mathbf{v}] = \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}'$ (Dot product)
6. $\dfrac{d}{dt}[\mathbf{u} \times \mathbf{v}] = \mathbf{u}' \times \mathbf{v} + \mathbf{u} \times \mathbf{v}'$ (Cross product)
7. $\dfrac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$ (Chain rule)

Proof of 5. Using the componentwise definition of the dot product:

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3,$$

we differentiate term by term:

$$\frac{d}{dt}(u_1 v_1 + u_2 v_2 + u_3 v_3) = u_1' v_1 + u_1 v_1' + u_2' v_2 + u_2 v_2' + u_3' v_3 + u_3 v_3' = \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}'.$$

Vector Functions of Constant Length

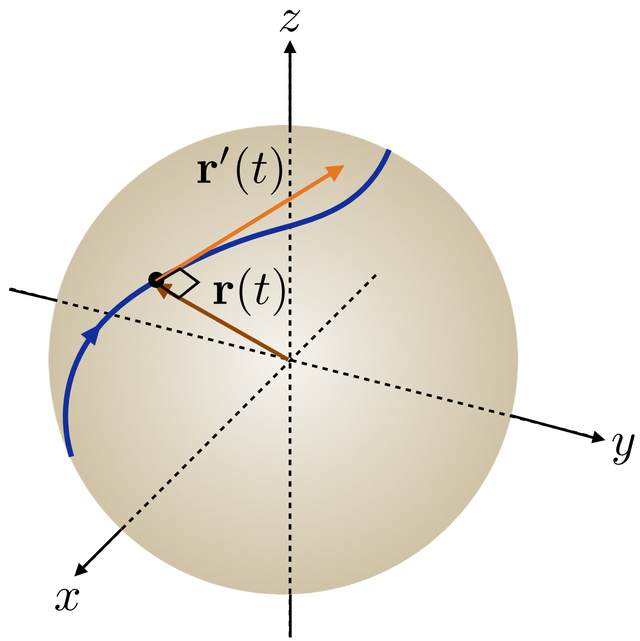
- For a particle moving on a sphere of radius c centered at the origin, the position vector $\mathbf{r}(t)$ satisfies

$$|\mathbf{r}(t)| = c \quad \text{(constant length)}.$$

- The velocity vector $\mathbf{v}(t) = \mathbf{r}'(t)$ is tangent to the path and tangent to the sphere, so it is perpendicular to $\mathbf{r}(t)$.

$$\begin{aligned}
\mathbf{r}(t) \cdot \mathbf{r}(t) &= c^2 && (|\mathbf{r}(t)| = c \text{ is constant}) \\
\implies \frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{r}(t)] &= 0 && \text{(differentiate both sides)} \\
\implies \mathbf{r}'(t) \cdot \mathbf{r}(t) + \mathbf{r}(t) \cdot \mathbf{r}'(t) &= 0 && \text{(using the dot product rule)} \\
\implies 2\mathbf{r}'(t) \cdot \mathbf{r}(t) &= 0
\end{aligned}$$

Hence, $\mathbf{r}'(t)$ and $\mathbf{r}(t)$ are **orthogonal**.



Integrals of Vector Functions

- A differentiable vector function $\mathbf{R}(t)$ is an **antiderivative** of $\mathbf{r}(t)$ on an interval I if

$$\frac{d\mathbf{R}}{dt} = \mathbf{r}(t) \quad \text{for all } t \in I.$$

- Every antiderivative of $\mathbf{r}(t)$ can be written as

$$\mathbf{R}(t) + \mathbf{C},$$

  where $\mathbf{C}$ is a constant vector.

- The set of all antiderivatives of $\mathbf{r}$ on I is called the **indefinite integral** of $\mathbf{r}$ on I:

$$\int \mathbf{r}(t)\, dt = \mathbf{R}(t) + \mathbf{C}.$$

Definition: Let $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$. If the components $f(t), g(t), h(t)$ are integrable over $[a, b]$, then $\mathbf{r}(t)$ is integrable over $[a, b]$, and its **definite integral** is

$$\int_a^b \mathbf{r}(t)\, dt = \left(\int_a^b f(t)\, dt \right) \mathbf{i} + \left(\int_a^b g(t)\, dt \right) \mathbf{j} + \left(\int_a^b h(t)\, dt \right) \mathbf{k}.$$

Integrals of Vector Functions

Example: Suppose we do not know the path of a hang glider, but only its acceleration vector

$$\mathbf{a}(t) = -(3\cos t)\mathbf{i} - (3\sin t)\mathbf{j} + 2\mathbf{k}.$$

We also know that initially (at time $t = 0$) the glider departed from the point $(3, 0, 0)$ with velocity $\mathbf{v}(0) = 3\mathbf{j}$. Find the glider's position as a function of t.

Solution. We have

$$\mathbf{a}(t) = -(3\cos t)\mathbf{i} - (3\sin t)\mathbf{j} + 2\mathbf{k},$$

$$\mathbf{v}(t) = \int \mathbf{a}(t)\,dt = -(3\sin t)\mathbf{i} + (3\cos t)\mathbf{j} + 2t\mathbf{k} + \mathbf{C}.$$

Initial condition $\mathbf{v}(0) = 3\mathbf{j}$ gives $\mathbf{C} = \mathbf{0}$, so $\mathbf{v}(t) = -(3\sin t)\mathbf{i} + (3\cos t)\mathbf{j} + 2t\mathbf{k}$.
Integrating again:

$$\mathbf{r}(t) = \int \mathbf{v}(t)dt = (3\cos t)\mathbf{i} + (3\sin t)\mathbf{j} + t^2\mathbf{k} + \mathbf{D}.$$

Using $\mathbf{r}(0) = (3, 0, 0)$ gives $\mathbf{D} = \mathbf{0}$, hence

$$\mathbf{r}(t) = (3\cos t)\mathbf{i} + (3\sin t)\mathbf{j} + t^2\mathbf{k}$$

is the glider's position function.

Arc Length in Space

Just as for plane curves, the length of a space curve can be computed from any convenient parametrization (proof omitted). Using $\mathbf{v} = \dfrac{d\mathbf{r}}{dt}$, the velocity vector, we can write:

$$L = \int_a^b |\mathbf{v}|\, dt.$$

Definition: The **length** of a smooth curve

$$\mathbf{r}(t) = x(t)\,\mathbf{i} + y(t)\,\mathbf{j} + z(t)\,\mathbf{k}, \quad a \le t \le b,$$

traced exactly once as t increases from a to b, is

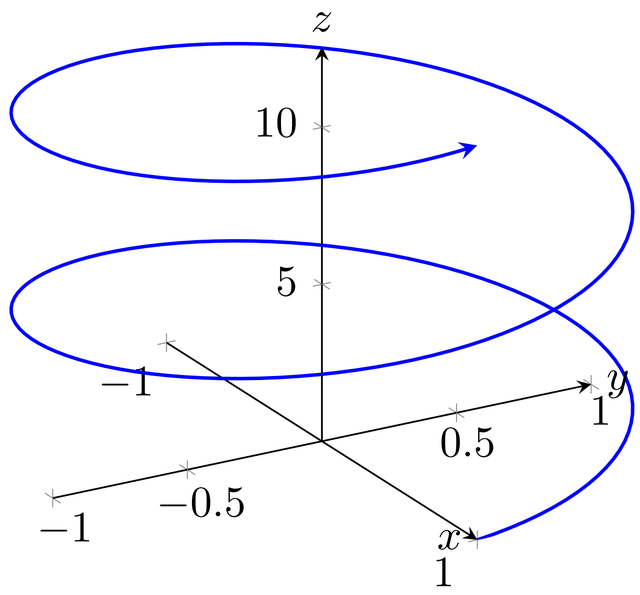
$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt.$$

Arc Length Parameter

Example: A glider is soaring upward along the helix

$$\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}.$$

How long is the glider's path from $t = 0$ to $t = 2\pi$?



Solution. The velocity is

$$\mathbf{r}'(t) = (-\sin t)\mathbf{i} + (\cos t)\mathbf{j} + \mathbf{k},$$

so the speed is

$$\left|\mathbf{r}'(t)\right| = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}.$$

Hence the length of the path is

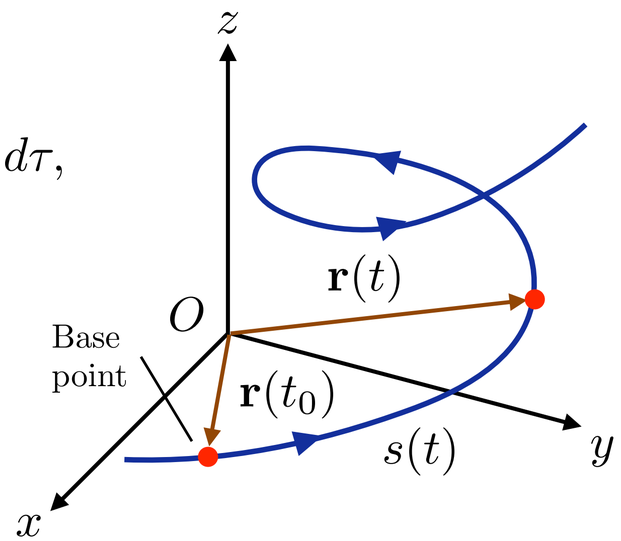
$$L = \int_0^{2\pi} \left|\mathbf{r}'(t)\right| dt = \int_0^{2\pi} \sqrt{2}\, dt = 2\pi\sqrt{2}.$$

Arc Length Parameter

For a smooth curve C with base point $P(t_0)$, each t determines a point $P(t) = (x(t), y(t), z(t))$ and a directed distance along the curve:

$$s(t) = \int_{t_0}^{t} |\mathbf{v}(\tau)|\, d\tau = \int_{t_0}^{t} \sqrt{[x'(\tau)]^2 + [y'(\tau)]^2 + [z'(\tau)]^2}\, d\tau,$$

where $\mathbf{v}(\tau) = \frac{d\mathbf{r}}{d\tau}$ is the velocity vector.



- If $t > t_0$, $s(t)$ is the distance from $P(t_0)$ to $P(t)$ along the curve.
- If $t < t_0$, $s(t)$ is negative.
- Each s determines a point on C, giving the curve a parametrization in terms of arc length.

Speed on a Smooth Curve: For a smooth curve, the arc length $s(t)$ is differentiable and its derivative is

$$\frac{ds}{dt} = |\mathbf{v}(t)|,$$

where $\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}$ is the velocity vector.

Unit Tangent Vector

If $s(t)$ is invertible, we can reparametrize the curve as $\mathbf{r}(s) = \mathbf{r}(t(s))$, linking points directly to their distance from $P(t_0)$.

Example: Consider the helix

$$\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}, \qquad t_0 = 0.$$

The arc length from t_0 to t is

$$s(t) = \int_{t_0}^{t} |\mathbf{v}(\tau)|\, d\tau = \int_{0}^{t} \sqrt{2}\, d\tau = \sqrt{2}\, t.$$

Solving for t gives $t = s/\sqrt{2}$. Substituting into $\mathbf{r}(t)$ yields the arc length parametrization of the helix:

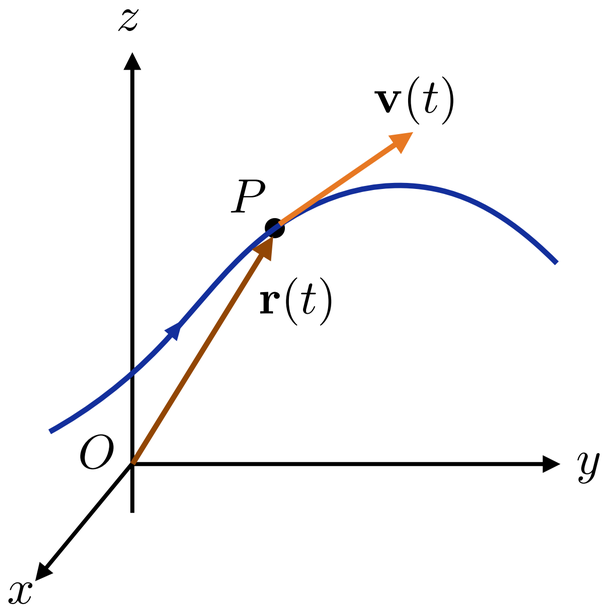
$$\mathbf{r}(s) = \cos\left(\frac{s}{\sqrt{2}}\right)\mathbf{i} + \sin\left(\frac{s}{\sqrt{2}}\right)\mathbf{j} + \frac{s}{\sqrt{2}}\,\mathbf{k}.$$

Unit Tangent Vector

- The velocity vector $\mathbf{v} = \dfrac{d\mathbf{r}}{dt}$ is tangent to the curve.
- The **unit tangent vector** is

$$\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|},$$

a differentiable vector whenever $\mathbf{v}$ is differentiable.



Example: Find the unit tangent vector of the curve

$$\mathbf{r}(t) = (3\cos t)\mathbf{i} + (3\sin t)\mathbf{j} + t^2\mathbf{k}$$

Solution. We have

$$\mathbf{v}(t) = \mathbf{r}'(t) = (-3\sin t)\mathbf{i} + (3\cos t)\mathbf{j} + (2t)\mathbf{k}$$

$$\implies |\mathbf{v}(t)| = \sqrt{(-3\sin t)^2 + (3\cos t)^2 + (2t)^2} = \sqrt{9 + 4t^2}.$$

Hence, the unit tangent vector is

$$\mathbf{T}(t) = \frac{\mathbf{v}(t)}{|\mathbf{v}(t)|} = \frac{-3\sin t}{\sqrt{9+4t^2}}\mathbf{i} + \frac{3\cos t}{\sqrt{9+4t^2}}\mathbf{j} + \frac{2t}{\sqrt{9+4t^2}}\mathbf{k}.$$